CS 410/510 · DATABASES · FALL 2026

Normalization Practice

First Normal Form, Second Normal Form, and Third Normal Form

Monday, November 2 · In-class practice · Elmasri & Navathe, Chapter 14

How to work each exercise. Every problem below yields to the same four steps. Use them even when the answer looks obvious — the habit is what carries you through the exam.

1. Write down the functional dependencies. If they are given, confirm you can read each one aloud as a sentence about the mini-world.
2. Find the candidate key by attribute closure. Check that no proper subset of it is also a key.
3. Name the violation precisely: a non-atomic attribute, a *partial* dependency on part of the key, or a *transitive* dependency through a non-prime attribute.
4. Decompose, then state what the decomposition bought you — which insert, update, or delete anomaly is now impossible.

WALKTHROUGH

Exercise 0 — Student advising records

Worked together at the start of class · target 1NF

The registrar keeps one row per student, in a spreadsheet exported to the following relation.

```
STUDENT_ADVISING(StudentID, StudentName, Address, Phones, Courses)
```

Two sample rows as currently stored.

StudentID	StudentName	Address	Phones	Courses
S1042	Marisol Vega	1910 University Dr, Boise, ID 83725	{208-426-1000, 208-426-5555}	{(CS 410, Fall 2026, A), (CS 321, Spring 2026, B)}
S1198	Theo Blackfoot	455 Lusk St, Boise, ID 83706	{208-426-3317}	{(CS 410, Fall 2026, B)}

1. Identify every reason this relation fails First Normal Form. There are three distinct reasons, and they are not all the same kind of problem.
2. Elmasri and Navathe give three standard remedies for a multivalued attribute. State all three and say what each costs.
3. Produce a 1NF schema. Mark the primary key of every relation and every foreign key.

Work space

1NF Exercise 1 — Senior design projects

About 8 minutes

```
STUDENT_PROJECT(ProjectID, ProjectTitle, Sponsor, SponsorContact,
TeamMembers, Milestones)
```

One sample row as currently stored.

ProjectID	ProjectTitle	Sponsor	SponsorContact	TeamMembers	Milestones
P-17	Campus Energy Dashboard	Idaho Power	Dana Ruiz / druiz@idahopower.com / 208-555-0144	{S1042, S1198, S1307}	{(Proposal, 2026-09-15), (Prototype, 2026-10-20), (Demo, 2026-12-02)}

1. Name each attribute that violates 1NF and classify it: composite, multivalued, or a nested relation.
2. Decompose into 1NF. Mark all primary and foreign keys.
3. TeamMembers and Milestones are both multivalued. Explain why they must go into two separate relations rather than one.

Work space

1NF Exercise 2 — Advisor load

About 8 minutes · read the question carefully

```
ADVISOR_LOAD(AdvisorID, AdvisorName,
             Advisee1, Advisee1Major,
             Advisee2, Advisee2Major,
             Advisee3, Advisee3Major)
```

Two sample rows as currently stored.

AdvisorID	AdvisorName	Advisee1	Advisee1Major	Advisee2	Advisee2Major	Advisee3	Advisee3Major
A-08	Cutchin	S1042	CS	S1198	CS	S1307	Math
A-12	Nakamura	S1455	CS	NULL	NULL	NULL	NULL

1. Is this relation in First Normal Form? Answer yes or no, and defend the answer against the formal definition — not against your instinct.
2. Whatever your answer to (1), this is a poor design. Give four specific problems it causes. At least one should be about writing queries.
3. Fix it. Your fix should not introduce a bridge table; explain why one is not needed here.

Work space

2NF Exercise 3 — Enrollment

About 10 minutes

```
ENROLLMENT(StudentID, CourseNum, StudentName, StudentMajor,
CourseTitle, Credits, Term, Grade)
```

FUNCTIONAL DEPENDENCIES

F1 `StudentID → StudentName, StudentMajor`

F2 `CourseNum → CourseTitle, Credits`

F3 `{StudentID, CourseNum} → Term, Grade`

1. Find the candidate key. Show the attribute closure that proves it, and show why neither attribute alone is a key.
2. Is the relation in 2NF? If not, list every partial dependency by name.
3. Give one insert anomaly, one update anomaly, and one delete anomaly this schema permits. Be concrete — use the mini-world, not the abstract definitions.
4. Decompose into 2NF. Mark all keys.

Work space

2NF Exercise 4 — Section roster

About 10 minutes · harder than Exercise 3

```
SECTION_ROSTER(CourseNum, SectionNum, Semester, CourseTitle, Credits,
               InstructorID, InstructorName, Room, RoomCapacity,
SeatsTaken)
```

FUNCTIONAL DEPENDENCIES

F1 `CourseNum → CourseTitle, Credits`

F2 `{CourseNum, SectionNum, Semester} → InstructorID, Room, SeatsTaken`

F3 `InstructorID → InstructorName`

F4 `Room → RoomCapacity`

1. Find the candidate key. Show the closure, and show that no two-attribute subset of it is a key.
2. Exactly one of these dependencies is a partial dependency. Which one? For each of the other three, say in one sentence why it is not partial.
3. Decompose into 2NF — 2NF only. Do not go further.
4. Your result still violates 3NF. Say which dependencies are responsible. You will fix them in Exercise 5.

Work space

3NF Exercise 5 — Section roster, continued

About 12 minutes

The relation below is already in 2NF. It is the Exercise 4 result with one attribute added, and it stands on its own — you do not need your Exercise 4 answer to work it.

```
SECTION_ROSTER(CourseNum, SectionNum, Semester, InstructorID,
InstructorName,
               InstructorDept, Room, RoomCapacity, SeatsTaken)
```

FUNCTIONAL DEPENDENCIES

F2 `{CourseNum, SectionNum, Semester} → InstructorID, Room, SeatsTaken`

F3 `InstructorID → InstructorName, InstructorDept`

F4 `Room → RoomCapacity`

1. Confirm the relation is in 2NF but not 3NF. Write out each transitive dependency in the form `key → X → Y`.
2. Give one insert anomaly and one delete anomaly that follow from the Room dependency specifically.
3. Decompose into 3NF. Mark every primary key and every foreign key.
4. Argue that your decomposition is lossless: name the join attribute for each pair and say why it is enough.
5. Is every original dependency still enforceable on a single relation? Name the relation that enforces each one.

Work space